Chapter 18, Problem 30P

Problem

For a linear system with input $x(t)$ and output $y(t)$, find the impulse response for the following cases:

(a) $x(t) = e{-}atu(t)$, $y(t) = u(t) - u(-t)$

(b) $x(t) = e{-}tu(t)$, $y(t) = e{-}2tu(t)$

(c) $x(t) = \delta(t)$, $y(t) = e{-}at \sin btu(t)$

Step-by-step solution

Step 1 of 3

(A) $x(t) = e^{-at}u(t) \Rightarrow X(\omega) = \dfrac{1}{j\omega + a}$

$$y(t) = u(t) - u(-t) \Rightarrow Y(\omega) = \frac{2}{j\omega}$$

$$H(\omega)\frac{Y(\omega)}{X(\omega)} = \frac{\left(\dfrac{2}{j\omega}\right)}{\left(\dfrac{1}{j\omega + a}\right)} = \frac{2(j\omega + a)}{j\omega} = 2 + \frac{2a}{j\omega}$$

$$\Rightarrow h(t) = 2\delta(t) + \frac{2a}{2}\operatorname{sgn}(t)$$

$$\boxed{h(t) = 2\delta(t) + a\operatorname{sgn}(t)}$$

Step 2 of 3

(B) $x(t) = e^{-t}u(t) \Rightarrow X(\omega) = \dfrac{1}{1 + j\omega}$

$$y(t) = e^{-2t}u(t) \Rightarrow Y(\omega) = \frac{1}{2 + j\omega}$$

$$\Rightarrow H(\omega) = \frac{1 + j\omega}{2 + j\omega} = -\frac{2 + j\omega - 1}{2 + j\omega} = 1 - \frac{1}{j\omega + 2} \qquad \left[H(\omega) = \frac{Y(\omega)}{X(\omega)}\right]$$

$$\Rightarrow \boxed{h(t) = \delta(t) - e^{-2t}u(t)}$$

Step 3 of 3

(C) $x(t) = \delta(t) \qquad \Rightarrow X(\omega) = 1$

$$y(t) = e^{-at}\sin bt \;\; u(t) \qquad \Rightarrow Y(\omega) = \frac{b}{(a + j\omega)^2 + b^2}$$

$$H(\omega) = \frac{Y(\omega)}{X(\omega)} = \frac{b}{(a + j\omega)^2 + b^2}$$

$or,\qquad \boxed{h(t) = e^{-at}\sin bt \;\; u(t)}$